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Java SE 27 & JDK 27

Interface GarbageCollectorMXBean

All Superinterfaces:

`MemoryManagerMXBean`, `PlatformManagedObject`

All Known Subinterfaces:

`GarbageCollectorMXBean`

`public interface` **`GarbageCollectorMXBean`**
`extends` `MemoryManagerMXBean`

The management interface for the garbage collection of the Java virtual machine. Garbage collection is the process that the Java virtual machine uses to find and reclaim unreachable objects to free up memory space. A garbage collector is one type of `memory manager`.

A Java virtual machine may have one or more instances of the implementation class of this interface. An instance implementing this interface is an MXBean that can be obtained by calling the `ManagementFactory.getGarbageCollectorMXBeans()` method or from the `platform MBeanServer` method.

The `ObjectName` for uniquely identifying the MXBean for a garbage collector within an MBeanServer is:

> `java.lang:type=GarbageCollector`,`name=`*collector's name*

It can be obtained by calling the `PlatformManagedObject.getObjectName()` method. A platform usually includes additional platform-dependent information specific to a garbage collection algorithm for monitoring.

Since:

1.5

See Also:

`ManagementFactory.getPlatformMXBeans(Class)`,
`MemoryMXBean`,
JMX Specification.,
Ways to Access MXBeans

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
`long`	**`getCollectionCount`**`()`	

		Returns the total number of collections that have occurred.
long	getCollectionTime()	Returns the approximate accumulated collection elapsed time in milliseconds.

Methods declared in interface MemoryManagerMXBean

getMemoryPoolNames, getName, isValid

Methods declared in interface PlatformManagedObject

getObjectName

Method Details

getCollectionCount

```
long getCollectionCount()
```

Returns the total number of collections that have occurred. This method returns `-1` if the collection count is undefined for this collector.

Returns:

the total number of collections that have occurred.

getCollectionTime

```
long getCollectionTime()
```

Returns the approximate accumulated collection elapsed time in milliseconds. This method returns `-1` if the collection elapsed time is undefined for this collector.

The Java virtual machine implementation may use a high resolution timer to measure the elapsed time. This method may return the same value even if the collection count has been incremented if the collection elapsed time is very short.

Returns:

the approximate accumulated collection elapsed time in milliseconds.

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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